MH2500 Probability and Introduction to Statistics

Slides for Week 5-1

Topics: Geometric Distribution, Hypergeometric Distribution
Sums of Random Variables

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Geometric Distribution

The **geometric distribution** is also constructed from independent Bernoulli trials, but from an infinite sequence.

- On each trial, a success occurs with probability p.

Let X be the total number of trials up to and including the first success.

- If $X = k$, then there must be $k-1$ failures preceding the first success.
- The probability is $P\{X = k\} = (1-p)^{k-1}p, \qquad k = 1, 2, \ldots.$

- The sum of probabilities is 1 because

$$\sum_{k=1}^{\infty} P\{X = k\} \;=\; \sum_{k=1}^{\infty}(p(1-p)^{k-1}) \;=\; p\sum_{j=0}^{\infty}(1-p)^j = p \cdot \frac{1}{1-(1-p)}.$$

Question

Let $X \sim \text{Geometric}(p)$. Find $\mathbb{E}(X)$ and $\text{Var}(X)$.

Let $q = 1 - p$. Then

$$\mathbb{E}(X) = \sum_{k=1}^{\infty} kq^{k-1}p = p\sum_{k=1}^{\infty} kq^{k-1}.$$

Let $x = \sum_{k=1}^{\infty} kq^{k-1} = 1 + 2q + 3q^2 + \cdots + kq^{k-1} + \cdots$. Then

$$qx = \sum_{k=1}^{\infty} kq^k = q + 2q^2 + \cdots + (k-1)q^{k-1} + \cdots,$$

and

$$x - qx = 1 + q + q^2 + \cdots + q^{k-1} + \cdots = \frac{1}{1-q}$$

$$x = \frac{1}{(1-q)^2}.$$

This gives

$$\mathbb{E}(X) = p\frac{1}{(1-q)^2} = p\,\frac{1}{p^2} = \frac{1}{p}.$$

Calculating $\mathbb{E}(X(X-1))$

$$\mathbb{E}(X(X-1)) = \sum_{k=1}^{\infty} k(k-1)q^{k-1}p = pq\sum_{k=2}^{\infty} k(k-1)q^{k-2}$$

As $\left(\sum_{k=0}^{\infty} q^k\right)'' = 2\cdot 1 + 3\cdot 2q + 4\cdot q^2 + \cdots + k(k-1)q^{k-2} + \cdots$, we have

$$\sum_{k=2}^{\infty} k(k-1)q^{k-2} = \left(\sum_{k=0}^{\infty} q^k\right)'' = \left(\frac{1}{1-q}\right)'' = \left(\left(\frac{1}{1-q}\right)'\right)'$$

$$= \left(\frac{1}{(1-q)^2}\right)' = \frac{2}{(1-q)^3} = \frac{2}{p^3}.$$

Thus,

$$\mathbb{E}(X(X-1)) = pq\frac{2}{p^3} = \frac{2q}{p^2}.$$

$$
\begin{aligned}
\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}\,X)^2 &= \mathbb{E}(X(X-1)) + (\mathbb{E}\,X) - (\mathbb{E}\,X)^2 \\
&= \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q+p-1}{p^2} \\
&= \frac{1-p}{p^2}.
\end{aligned}
$$

Hypergeometric Distribution

Suppose an urn contains N balls, of which m are white and $N - m$ are black. Let X denote the number of white balls drawn when taking n balls *without replacement*. Then

$$P(X = k) = \frac{\binom{m}{k}\binom{N-m}{n-k}}{\binom{N}{n}}.$$

- X is a **hypergeometric random variable** with parameters n, N, and m, shorthanded as $X \sim \text{Hypergeometric}(n, N, m)$.

Comparison with Binomial Distribution

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- With replacement:
- Without replacement:

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Indeed, when $X \sim \text{Hypergeometric}(n, N, m)$,

$$\mathbb{E}(X) = n \cdot \frac{m}{N}, \quad \text{Var}(X) = n \cdot \frac{m}{N}\left(1 - \frac{m}{N}\right)\left(1 - \frac{n-1}{N-1}\right).$$

Expectation of Sums of Random Variables

Recall that a random variable X is a function $X : \Omega \to \mathbb{R}$ "that aggregates the outcomes $\omega \in \Omega$":

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Thus

$$\mathbb{E}(X) = \sum_x xP\{X = x\} = \sum_x x \sum_{\omega : X(\omega) = x} P(\omega) = \sum_x \left(\sum_{\omega : X(\omega) = x} (xP(\omega)) \right)$$

$$= \sum_x \left(\sum_{\omega : X(\omega) = x} (X(\omega)P(\omega)) \right) = \sum_{\omega \in \Omega} X(\omega)P(\omega)$$

Theorem

Suppose that X, Y are random variables on the same sample space Ω. Then $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$.

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Proof:

$$\begin{aligned}\mathbb{E}(X+Y) &= \sum_{\omega\in\Omega}(X(\omega)+Y(\omega))P(\omega) \\ &= \sum_{\omega} X(\omega)P(\omega) + \sum_{\omega} Y(\omega)P(\omega) = \mathbb{E}(X) + \mathbb{E}(Y).\end{aligned}$$

Corollary

Let $X_1, \ldots, X_n$ be random variables on the same sample space Ω, then

$$\mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n).$$

Binomial Distribution: Revisited

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$$\mathbb{E}(X) = \mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n) = np.$$